

Rohan Sen

Department of Finance
The Hong Kong University of Science and Technology

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PROFESSIONAL EXPERIENCE

The Hong University of Science and Technology

Post-doctoral fellow

Advisor: [Yingying Li](#)

Hong Kong

March 2026 – present

HCL Technologies

Senior Software Engineer

Bangalore, India

September 2019 – December 2020

RESEARCH INTERESTS

- Kernel methods
- Financial econometrics
- Statistical learning
- Explainability

EDUCATION

Università della Svizzera italiana

Ph.D. in Econometrics and Statistics

Thesis: [On applications of kernel methods to financial econometrics](#)

Main advisor: [Paul Schneider](#)

Co-advisor: [Michael Multerer](#)

Lugano, Switzerland

January 2021 – January 2026

Indian Institute of Technology Guwahati

M.Sc. in Mathematics and Computing

Thesis: Stein's method and its applications

Project advisor: [Subhamay Saha](#)

Guwahati, India

July 2017 – June 2019

Ramakrishna Mission Vidyamandira

B.Sc. in Mathematics

Minors in Statistics and Computer Science

Howrah, India

July 2014 – July 2017

ACADEMIC APPOINTMENTS

Stanford University

Visiting Student Researcher

Host: [Markus Pelger](#)

California, USA

September 2025 – December 2025

PUBLICATIONS AND PREPRINTS

- *Kernel-based nonparametric tests for shape constraints*; working paper, October 2025.
- *Learning the stochastic discount factor via nonparametric option portfolios*, with E. Luzzi and P. Schneider; working paper, October 2025.
- *Probabilistic energy forecasting through quantile regression in reproducing kernel Hilbert spaces*, with L. Pernigo and D. Baroli; **ACM SIGENERGY Energy Informatics Review**, February 2025.
- *Fast empirical scenarios*, with M. Multerer and P. Schneider; **Journal of Computational Mathematics and Data Science**, September 2024.
- *Observation-specific explanations through scattered data approximation*, with V. Ghidini, M. Multerer, and J. Quizi; **Explainable Artificial Intelligence, XAI 2024, Communications in Computer and Information Science**, July 2024.

CONFERENCES AND CONTRIBUTED TALKS

- 19th International Joint Conference on Computational and Financial Econometrics (CFE) and Computational and Methodological Statistics (CMStatistics), Birkbeck, University of London, UK, December 2025; *Kernel-based nonparametric tests for shape constraints*.
- 2nd IAS-SBM Joint Workshop on Financial Econometrics in the Big Data Era, Hong Kong University of Science and Technology, Hong Kong, August 2025; *Optimal variance swaps*.
- 17th Annual Conference of SoFiE (Society for Financial Econometrics), ESSEC Business School, France, June 2025.
- 18th International Joint Conference on Computational and Financial Econometrics (CFE) and Computational and Methodological Statistics (CMStatistics), King's College, London, UK, December 2024.
- FinEML (Financial Econometrics Meets Machine Learning) Conference, USI Lugano, Switzerland, November 2024.
- 2nd World Conference on XAI (eXplainable Artificial Intelligence), Valletta, Malta, July 2024.
- FinEML (Financial Econometrics Meets Machine Learning) Conference, Erasmus University, Rotterdam, Netherlands, November 2023; *Fast empirical scenarios*.
- 15th Annual Conference of SoFiE (Society for Financial Econometrics), Seoul, South Korea, June 2023; *Fast empirical scenarios*.
- Conference on Stochastic Optimal Control in Economics, Finance, and Learning Theory, ETH Zürich, Switzerland, June 2023.
- 2nd Conference of Young Applied Mathematicians, Arenzano, Italy, September 2022; *Kernel-based method for large empirical truncated moment problem*.
- SFI (Swiss Finance Institute) Research Days, Gerzensee, Switzerland, June 2022.

- Conference on Probability and Stochastic Processes, Indian Statistical Institute, Kolkata, India, March 2022 (attended online).

AWARDS AND HONORS

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| • USI Doctoral Mobility Grant
<i>Funded a semester visit to Stanford University</i> | 2025 |
| • CSIR NET-JRF (All India Rank 42)
<i>National fellowship for Ph.D. studies in India</i> | 2019 |
| • Joint Admission Test for Masters (All India Rank 155) | 2017 |
| • National Inspire Scholarship (Top 1% in 12th Board Examination)
<i>National fellowship offered for five years after high school</i> | 2014 |

TEACHING EXPERIENCE

Università della Svizzera italiana
Teaching Assistant, *Numerical Methods*

Lugano, Switzerland
Spring 2022, 2023, 2024, 2025

- Delivered problem-solving sessions for students in Bachelor in Economics (Quantitative Methods) and Master in Finance (Quantitative Finance).
- Prepared and graded assignments, exams, and project work.
- Provided one-on-one student consultation and feedback.

PROFESSIONAL TRAINING AND WORKSHOPS

- Workshop on Advanced C++, CSCS Swiss National Supercomputing Centre, Switzerland, October 2023.
- Workshop on Introduction to Decision Making and Uncertainty, Institute for Mathematical and Statistical Innovation, July 2021 (held online).

SKILLS

- **Programming:** C++, Python, MATLAB, R.
- **Languages:** English (fluent), Italian (intermediate), Bengali (native), Hindi (intermediate).